

I. DATA INPUT

Locations Dataset:

```
{'id': 0, 'name': 'JNE Cabang Utama Bandung', 'weight': 0.0}
{'id': 1, 'name': 'Gacoan Bojongsoang', 'weight': 3.0}
{'id': 2, 'name': 'Bubur Ayam Yuyu Batununggal', 'weight': 0.5}
{'id': 3, 'name': 'Warung Nasi Sunda Mang Ujang', 'weight': 5.0}
{'id': 4, 'name': 'Kost 22', 'weight': 2.0}
{'id': 5, 'name': 'Komplek Taman Cibaduyut Indah', 'weight': 1.5}
{'id': 6, 'name': 'Kostan Pondok Zeta', 'weight': 3.0}
{'id': 7, 'name': 'Griya Bali - Rumah Kos', 'weight': 4.3}
{'id': 8, 'name': 'Rumah Jahit Neniyouna Fashion', 'weight': 4.2}
{'id': 9, 'name': "Tamara's House", 'weight': 6.0}
{'id': 10, 'name': "Produsen Gamis Syar'i Aqiefa Alfa", 'weight': 5.0}
{'id': 11, 'name': 'Sukamenak Indah Blok K51a', 'weight': 3.9}
{'id': 12, 'name': 'Kosan Rani Kazool', 'weight': 5.7}
{'id': 13, 'name': "Sim Simo House's", 'weight': 4.8}
```

Distance Matrix Dataset:

```
[0.0, 4.9, 2.2, 7.0, 5.7, 4.4, 3.6, 2.4, 1.5, 8.1, 6.6, 7.3, 5.4, 2.9]
[4.9, 0.0, 3.2, 7.1, 1.3, 5.9, 1.0, 5.7, 3.7, 7.8, 9.8, 9.0, 7.9, 4.5]
[2.2, 3.2, 0.0, 8.0, 4.0, 5.4, 3.1, 3.0, 0.9, 7.2, 7.6, 8.4, 7.0, 3.1]
[7.0, 7.1, 8.0, 0.0, 8.3, 2.7, 7.3, 6.8, 7.7, 1.4, 3.1, 2.3, 9.0, 9.3]
[5.7, 1.3, 4.0, 8.3, 0.0, 3.9, 1.8, 6.5, 4.6, 9.0, 11.0, 10.0, 8.7, 5.3]
[4.4, 5.9, 5.4, 2.7, 3.9, 0.0, 4.9, 5.1, 6.0, 2.0, 3.0, 3.6, 7.9, 7.6]
[3.6, 1.0, 3.1, 7.3, 1.8, 4.9, 0.0, 6.3, 3.5, 6.8, 7.2, 8.0, 8.5, 4.8]
[2.4, 5.7, 3.0, 6.8, 6.5, 5.1, 6.3, 0.0, 2.8, 5.3, 6.2, 6.8, 4.3, 3.0]
[1.5, 3.7, 0.9, 7.7, 4.6, 6.0, 3.5, 2.8, 0.0, 6.7, 7.5, 8.2, 6.3, 2.4]
[8.1, 7.8, 6.7, 1.4, 8.1, 2.0, 6.3, 5.3, 6.7, 0.0, 2.7, 2.1, 9.6, 8.3]
[6.6, 9.8, 7.6, 3.1, 9.0, 3.0, 7.2, 6.2, 7.5, 2.7, 0.0, 0.95, 8.2, 8.5]
[7.3, 8.9, 8.4, 2.3, 9.6, 3.6, 7.8, 6.8, 8.2, 2.3, 0.95, 0.0, 8.9, 9.6]
[5.4, 7.9, 6.0, 9.0, 8.7, 7.9, 8.5, 4.3, 5.3, 10.0, 8.7, 9.4, 0.0, 3.9]
[2.9, 4.5, 3.1, 9.3, 5.3, 7.6, 4.8, 3.0, 2.4, 10.0, 9.0, 9.6, 4.1, 0.0]
```

II. ALGORITHMS SIMULATION

1. Subsidy - Harga BBM: Rp 5000/liter

1a. Basic Heuristic Algorithm (Greedy NN)

Route:

JNE Cabang Utama Bandung -> Rumah Jahit Neniyouna Fashion -> Bubur Ayam Yuyu Batununggal -> Griya Bali - Rumah Kos -> Sim Simo House's -> Kosan Rani Kazool -> Gacoan Bojongsoang -> Kostan Pondok Zeta -> Kost 22 -> Komplek Taman Cibaduyut Indah -> Tamara's House -> Warung Nasi Sunda Mang Ujang -> Sukamenak Indah Blok K51a -> Produsen Gamis Syar'i Aqiefa Alfa -> JNE Cabang Utama Bandung

- Performance Metrics -

Total Distance: 40.35 km

Execution Time: 0.00005260 sec

Fuel Used: 1.4073 liter

- Cost Breakdown -

Fuel Cost: Rp 7,036.56

Server Cost: Rp 0.00

TCO: Rp 7,036.57

1b. Algoritma Eksak (Held-Karp DP)

Route:

JNE Cabang Utama Bandung -> Rumah Jahit Neniyouna Fashion -> Bubur Ayam Yuyu Batununggal -> Kostan Pondok Zeta -> Gacoan Bojongsoang -> Kost 22 -> Komplek Taman Cibaduyut Indah -> Tamara's House -> Warung Nasi Sunda Mang Ujang -> Sukamenak Indah Blok K51a -> Produsen Gamis Syar'i Aqiefa Alfa -> Griya Bali - Rumah Kos -> Kosan Rani Kazool -> Sim Simo House's -> JNE Cabang Utama Bandung

- Performance Metrics -

Total Distance: 35.65 km

Execution Time: 0.30037180 sec

Fuel Used: 1.2168 liter

- Cost Breakdown -

Fuel Cost: Rp 6,083.93

Server Cost: Rp 15.02

TCO: Rp 6,098.94

--- PERBANDINGAN ---

Heuristic TCO: Rp 7,036.57

Exact TCO: Rp 6,098.94

Recommendation:

Use Exact

2. Crisis - Harga BBM: Rp 20000/liter

2a. Basic Heuristic Algorithm (Greedy NN)

Route:

JNE Cabang Utama Bandung -> Rumah Jahit Neniyouna Fashion -> Bubur Ayam Yuyu Batununggal -> Griya Bali - Rumah Kos -> Sim Simo House's -> Kosan Rani Kazool -> Gacoan Bojongsoang -> Kostan Pondok Zeta -> Kost 22 -> Komplek Taman Cibaduyut Indah -> Tamara's House -> Warung Nasi Sunda Mang Ujang -> Sukamenak Indah Blok K51a -> Produsen Gamis Syar'i Aqiefa Alfa -> JNE Cabang Utama Bandung

- Performance Metrics -

Total Distance: 40.35 km

Execution Time: 0.00005620 sec

Fuel Used: 1.4073 liter

- Cost Breakdown -

Fuel Cost: Rp 28,146.26

Server Cost: Rp 0.00

TCO: Rp 28,146.26

2b. Algoritma Eksak (Held-Karp DP)

Route:

JNE Cabang Utama Bandung -> Rumah Jahit Neniyouna Fashion -> Bubur Ayam Yuyu Batununggal -> Kostan Pondok Zeta -> Gacoan Bojongsoang -> Kost 22 -> Komplek Taman Cibaduyut Indah -> Tamara's House -> Warung Nasi Sunda Mang Ujang -> Sukamenak Indah Blok K51a -> Produsen Gamis Syar'i Aqiefa Alfa -> Griya Bali - Rumah Kos -> Kosan Rani Kazool -> Sim Simo House's -> JNE Cabang Utama Bandung

- Performance Metrics -

Total Distance: 35.65 km

Execution Time: 0.38419070 sec

Fuel Used: 1.2168 liter

- Cost Breakdown -

Fuel Cost: Rp 24,335.71

Server Cost: Rp 19.21

TCO: Rp 24,354.92

--- PERBANDINGAN ---

Heuristic TCO: Rp 28,146.26

Exact TCO: Rp 24,354.92

Recommendation:

Use Exact

Confirmation:

All graphic visualizations were successfully saved to the docs/ folder